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Schrödinger vs. the Relativistic TauTimp Ondulatory Equation: Comparative Synthesis for H, He, Li, Be, B, C

Sources (all in this directory): electron_TauTimp_final_RO.md (TauTimp ondulatory equation and its relativistic extension, §6.3), Corectie_Relativista_TauTimp_v2_RO.md (rigorous relativistic kinetic correction $\hat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$, numerically validated for all 13 states of the series), Precizie_Schrodinger_TauTimp_EN.md (accuracy against NIST spectral/ionization data), and the six per-element Heisenberg-equilibrium documents: Hidrogen_Echilibru_S1_RO, Heliu_Echilibru_2electroni_EN, Litiu_Echilibru_TauTimp_RO, Beriliu_Echilibru_TauTimp_RO, Bor_Echilibru_TauTimp_RO, Carbon_Echilibru_TauTimp_RO

Experimental reference: NIST Atomic Spectra Database, NIST Ground Levels and Ionization Energies

Energy quantum absorbed per electron: $\Delta E = 3.59$ eV

Date: August 23, 2026

1 Purpose

This document consolidates, into a single English synthesis, the percentage differences reported across the directory's H–C document series between two theoretical predictions for each electron's equilibrium radius/energy — the standard **Schrödinger** equation and the **TauTimp ondulatory equation**, including its rigorous relativistic kinetic extension, $\hat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$, already present in the TauTimp source document (§6.3) — for the first six elements of the periodic table: **H, He, Li, Be, B, C**. It draws no new physics; every number below is taken from, or directly derived from, the files listed above.

Three logically distinct percentage gaps appear in the source series, and this synthesis keeps them separate rather than conflating them:

1. **Schrödinger vs. TauTimp** (radius/energy, same Z_{eff}, same ΔE) — tests whether the two *wave equations* differ at atomic scale.
2. **Relativistic correction vs. non-relativistic** (same equation, before/after adding $\hat{H}_{\text{rel}}$) — tests the size of relativistic kinetic effects, common to both equations.
3. **Model prediction vs. NIST experiment** — tests the accuracy of the underlying *atomic model* (screened, independent-electron hydrogenic approximation), independent of which wave equation computed it.

2 Methodology recap

For each element, every electron is treated as an independent hydrogenoid particle in a screened Coulomb potential $-Z_{eff}ke^2/r$ (Slater screening for H–Be, Clementi–Raimondi for B–C), starting at the natural equilibrium $\beta_0 = a_0/Z_{eff}$, $E_0 = -13.6057\ Z_{eff}^2/n^2$ eV. After each electron absorbs $\Delta E = 3.59$ eV, the new equilibrium radius r' is obtained in closed form from $\langle H\rangle(r') = E_0 + \Delta E$, selecting the branch in which the electron moves closer to the nucleus, subject to Heisenberg saturation $\Delta x \Delta p = \hbar$ ($1s$, no radial node) or $\hbar/2$ ($2s$, $2p$, with a node/angular momentum).

The TauTimp ondulatory equation (source document, eq. 6) reduces algebraically to the same functional form as Schrödinger at atomic scale, with an additive cosmological-curvature tail of order $(r/R)^4$, $R = 4.2355 \times 10^{22}$ m. The source document's own numerical validation table (§4.3) confirms this identity to **machine precision** for every node-free state ($l = n - 1$: $1s$, $2p$, $3d$, …) and to a conservative bound $\lesssim 1.65 \times 10^{-10}$ (relative) for node-bearing states ($2s$, tested there only by analogy with $2p$, since $2s$ itself is not in the source's explicit validation range). The relativistic kinetic term $\hat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$ is already part of the *extended* TauTimp equation (§6.3) and is numerically identical to the correction applied on the Schrödinger side — it is not a separate, competing prediction.

3 Table 1 — Equilibrium radius: Schrödinger vs. TauTimp (% difference)

$\beta_0 = a_0/Z_{eff}$ (initial equilibrium); r' = radius after absorbing $\Delta E = 3.59$ eV (physical branch, electron moves inward). Both equations give the *same* r' — the "% difference" column is therefore a **verification of equivalence**, not a measured discrepancy.

Element	State	Z_{eff}	β_0 (pm)	$r'_{\text{Schrödinger}} = r'_{\text{TauTimp}}$ (pm)	Schrödinger–TauTimp % difference	Validation basis
H	$1s$	1.000	52.9177	34.9598	0% (machine precision)	source §4.3, row $l = 0, n =$

Element	State	Z_{eff}	β_0 (pm)	$r'_{\text{Schrödinger}} = r'_{\text{TauTimp}}$ (pm)	Schrödinger–TauTimp % difference	Validation basis
He	$1s$	1.700	31.1281	23.9049 (rel.: 23.9061)	$< 4 \times 10^{-131}\%$	1 multi-body Hartree extension, verified directly
Li	$1s$	2.700	19.5992	16.4664	$< 10^{-131}\%$	verified directly
Li	$2s$	1.300	40.7059	22.7374	$\lesssim 1.65 \times 10^{-8}\%$	conservative bound, by analogy with $2p$
Be	$1s$	3.700	14.3021	12.5586	$< 10^{-131}\%$	verified directly
Be	$2s$	1.950	27.1373	17.7735	$\lesssim 1.65 \times 10^{-8}\%$	conservative bound, by analogy with $2p$
B	$1s$	4.6795	11.3084*	10.1899	$< 10^{-132}\%$	verified directly
B	$2s$	2.580	20.5107*	14.6694	$\lesssim 1.65 \times 10^{-8}\%$	conservative bound, by analogy with $2p$
B	$2p$	2.420	21.8668*	15.3503	$\lesssim 1.65 \times 10^{-8}\%$	**verified directly** ($l = 1, n = 2$)
C	$1s$	5.6727	9.3285*	8.5539	$< 10^{-132}\%$	verified directly
C	$2s$	3.220	16.4341*	12.4590	$\lesssim 1.65 \times 10^{-8}\%$	conservative bound, by analogy with $2p$
C	$2p$	3.140	16.8528*	12.6982	$\lesssim 1.65 \times 10^{-8}\%$	**verified directly** ($l = 1, n = 2$)

*B/C initial radii β_0 computed from a_0/Z_{eff}, the same relation independently confirmed against the source-document values for H, He, Li, Be (agreement to 4–5 significant figures); not separately re-quoted in the B/C TauTimp documents, which report r' directly.

Reading this table: every entry shows agreement between Schrödinger and TauTimp far beyond any conceivable experimental resolution ($10^{-8}\%$ in the worst, conservative case for node-bearing $2s/2p$ states; effectively 0%, machine precision, for every node-free $1s/2p$ state directly validated in the source). The only source of *any* residual gap in the $2s$ row is that the TauTimp source document's own validation table never explicitly tests $l = 0, n = 2$ — the bound used is a deliberately conservative carry-over from the nearest validated state ($2p$), not a measured Schrödinger–TauTimp discrepancy.

4 Table 2 — Rigorous relativistic kinetic correction (identical for both equations)

$\Delta E_{\text{rigorous}} = -T_0^2/2m_e c^2 = -|E_0| \cdot 1/4\,(Z_{eff}\alpha/n)^2$, from $\hat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$ — the term already embedded in the extended TauTimp equation (§6.3); numerically identical whether attached to Schrödinger's or TauTimp's kinetic term, since both share the same non-relativistic T_0 at atomic scale (Table 1).

Element	State	E_0 (eV)	$\Delta E_{\text{rigorous}}$ (meV)	Correction as % of $\lvert E_0 \rvert$
H	$1s$	-13.6057	-0.181	0.00133%
He	$1s$	-39.3205	-1.513	0.00385%
Li	$1s$	-99.1855	-9.626	0.00971%
Li	$2s$	-5.7484	-0.032	0.00056%

Element	State	E_0 (eV)	$\Delta E_{\text{rigorous}}$ (meV)	Correction as % of $\lvert E_0 \rvert$
Be	$1s$	-186.2619	-33.947	0.01823%
Be	$2s$	-12.9339	-0.164	0.00127%
B	$1s$	-297.9337	-86.854	0.02915%
B	$2s$	-22.6412	-0.502	0.00222%
B	$2p$	-19.9201	-0.388	0.00195%
C	$1s$	-437.8247	-187.564	0.04284%
C	$2s$	-35.2673	-1.217	0.00345%
C	$2p$	-33.5367	-1.101	0.00328%

The correction grows monotonically with Z_{eff} (from 0.0013% at H-1s to 0.043% at C-1s) but never exceeds 0.05% across all thirteen states — **three to four orders of magnitude smaller than the model-vs-NIST gaps** in Table 3 below. (Note: the heuristic mass substitution $m_{\text{eff}} = m/\sqrt{1 - V^2/c^2}$ tested in Corectie_Relativista_TauTimp_v2_RO.md gives the correct sign but overestimates every value in this table by exactly a factor of 2; the rigorous $-\hat{p}^4/8m^3c^2$ values above are the ones to use quantitatively.)

5 Table 3 — Model accuracy vs. NIST experiment (independent of Schrödinger/TauTimp choice)

Reproduced from Precizie_Schrodinger_TauTimp_EN.md. Because Tables 1–2 establish that Schrödinger and (relativistic) TauTimp predict identical numbers, every entry below applies equally to both theories — the error against experiment is **not** a property of the wave equation.

Atom	Valence state	Z_{eff}	E_{model} (eV)	E^{NIST} (eV)	Model–NIST % error	Fraction explained by $\Delta E_{\text{rigorous}}$
H	$1s$	1.00	-13.5983	-13.5984	0.001%	2.5%
He^+	$1s$	2.00	-54.4154	-54.4178	0.004%	119.6%
He	$1s^2$	1.70	-78.6409	-79.0052	0.461%	0.10%
Li	$2s^1$	1.30	-5.7484	-5.3917	6.62%	0.009%
Be	$2s^2$	1.95	-12.9339	-9.3227	38.74%	0.005%
B	$2p^1$	2.42	-19.9201	-8.2980	140.06%	0.003%
C	$2p^2$	3.14	-33.5367	-11.2603	197.83%	0.005%

The error grows nearly exponentially with the number of $n = 2$-shell electrons, because the fixed-Z_{eff}, independent-electron model omits explicit electron–electron correlation — a limitation shared identically by Schrödinger and TauTimp, and left essentially untouched by the relativistic correction (which explains under 0.1% of any multi-electron gap).

6 Synthesis: the three-tier hierarchy of % differences

$$\underbrace{0\% \text{ to } 1.65 \times 10^{-8}\%}_{\text{Schrödinger vs. TauTimp (Table 1)}} \ll \underbrace{0.0013\% \text{ to } 0.043\%}_{\text{relativistic correction (Table 2)}} \ll \underbrace{0.001\% \text{ to } 198\%}_{\text{model vs. NIST reality (Table 3)}}$$

1. **Schrödinger and the relativistic TauTimp ondulatory equation are experimentally and numerically indistinguishable at atomic scale**, for every element H through C and every state tested ($1s$, $2s$, $2p$). The largest reported gap between them, $1.65 \times 10^{-8}\%$, is a conservative bound carried over from a state the TauTimp source document does not explicitly validate ($2s$) — not a measured physical discrepancy. Directly validated states ($1s$, $2p$) agree to machine precision.
2. **The rigorous relativistic kinetic correction** $\hat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$, already native to the extended TauTimp equation, is identical whether computed on the Schrödinger or the TauTimp side, and stays below 0.05% of E_0 across the whole series — real, monotonically growing with Z_{eff}, but three to four orders of magnitude smaller than the errors in Table 3.
3. **The real, physically significant percentage differences in this document series — up to** 198% **for carbon — come entirely from the level of approximation of the atomic model** (single screened effective charge, no explicit electron correlation), **not from any difference between the Schrödinger and TauTimp wave equations, and not from the relativistic kinetic correction.** This is the central, repeated finding across every per-element document in the directory (Precizie_Schrodinger_TauTimp_EN.md, and each *_TauTimp_RO.md file individually).

7 Conclusions

1. Across H, He, Li, Be, B, C, the relativistically-extended TauTimp ondulatory equation reproduces the Schrödinger equation's equilibrium radii and energies to within 0%–$1.65 \times 10^{-8}\%$ — effectively exact agreement at atomic scale, consistent with the TauTimp source document's own design (a cosmological-curvature correction of order $(r/R)^4 \lesssim 10^{-130}$, unmeasurable at any atomic distance).
2. The rigorous relativistic kinetic correction is small (≤0.05% of E_0) but real and common to both formulations; it should not be confused with, nor can it explain, the far larger discrepancies against NIST experimental data.

3. The percentage gaps that matter physically — 6.6% (Li) up to 198% (C) — reflect the fixed-effective-charge, independent-electron approximation shared by both theories, and can only be closed by explicit multi-electron correlation (Hartree–Fock, CI, coupled-cluster), not by choosing between Schrödinger and TauTimp, nor by refining the relativistic kinetic term further.

8 Appendix: constants used

Symbol	Meaning	Value
a_0	Bohr radius	52.9177 pm
$\hbar$	Reduced Planck constant	1.054572×10^{-34} J·s
α	Fine-structure constant	$7.2973525693 \times 10^{-3}$
$m_e c^2$	Electron rest energy	510,998.95 eV
R	TauTimp cosmological radius	4.2355×10^{22} m
ΔE	Energy quantum absorbed (each electron)	3.59 eV
Z_{eff}	Slater (H–Be) / Clementi–Raimondi (B–C) effective nuclear charge	per-state, Table 1

Precision of the Schrödinger and Ondulatory (TauTimp) Wave Equations Against NIST Spectral Data: Hydrogen, Helium, Lithium, Beryllium, Boron, Carbon

Sources: Heisenberg.docx (uncertainty principle), electron_TauTimp_final_RO.md (TauTimp ondulatory equation), Corectie_Relativista_TauTimp_v2_RO.md (relativistic kinetic correction, this conversation), the Hydrogen–Lithium–Beryllium–Boron–Carbon equilibrium series (Heisenberg equilibrium, this conversation)
Experimental data: NIST Atomic Spectra Database, NIST Ground Levels and Ionization Energies
Date: August 22, 2026

1 Introduction and central conclusion

This document compares the precision of **two** theoretical formulations — the standard Schrödinger equation and the ondulatory (TauTimp) wave equation from electron_TauTimp_final_RO.md — against real experimental data (NIST): ionization energies and spectral lines, for the first six atoms of the periodic table: H, He, Li, Be, B, C. Compared to the initial version, it now includes an explicit quantification of the relativistic kinetic correction with effective mass $m_{\text{eff}} = m/\sqrt{1 - V^2/c^2}$ (the correct, standard form of relativistic mass — see Corectie_Relativista_TauTimp_v2_RO.md), applied to all six elements.

The TauTimp source document already establishes (sections 4.2–4.3, numerically verified there and reconfirmed here for every atom in the series) that **the modified TauTimp ODE, with the cosmological radius R held constant, is algebraically identical to the Schrödinger equation at any atomic scale**, with a corrective cosmological tail of order $(r/R)^4 \lesssim 10^{-130}$ — completely unmeasurable. The consequence for this document:

> **The two theories are indistinguishable from one another when compared against NIST data.** Any error against experiment identified below is identical for Schrödinger and for TauTimp — neither is "more precise" at atomic scale. The real discrepancies from reality come from **the level of approximation of the atomic model used** (single hydrogenic electron vs. real multi-electron atom), not from the choice of wave equation.

2 Methodology

For **single-electron systems** (H, and the He^+ ion), the **exact** Schrödinger/TauTimp solution is used, with no screening approximation whatsoever — the comparison with NIST directly tests the accuracy of the wave equations themselves, including reduced-mass and relativistic-kinetic corrections.

For **neutral multi-electron atoms** (He, Li, Be, B, C), the screened hydrogenic model developed in the preceding documents of this series is used: Slater's rules for He/Li/Be, Clementi–Raimondi charges for B/C (more accurate, distinguishing $2s$ from $2p$). Here the comparison with NIST tests **both** the wave equation (identical for Schrödinger and TauTimp) **and** the quality of the independent-electron approximation with a fixed effective charge — the dominant source of error, as will become clear.

For the relativistic kinetic correction, we use the standard form of relativistic mass, $m_{\text{eff}}(V) = m/\sqrt{1 - V^2/c^2}$, inserted into the kinetic term of the energy functional (section "Quantifying the residual" details the derivation and

closed form), and compare the result to the rigorous relativistic term $\widehat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$, already present in the extended TauTimp ondulatory equation (source document, §6.3).

3 Hydrogen — near-perfect agreement

3.1 Ground-state energy: infinite mass vs. reduced mass

$$E_1(R_\infty) = -13.605693 \text{ eV} \qquad \text{(the } Z_{eff} = 1\text{, electron-mass convention used throughout the series)}$$

$$E_1(R_H) = -\mu c^2\alpha^2/2 = -13.598287 \text{ eV} \qquad \mu = \frac{m_e m_p}{m_e + m_p} \text{ (electron–proton reduced mass)}$$

$$E_1^{\text{NIST}} = -13.598434 \text{ eV}$$

Model	Relative error vs. NIST
R_∞ (infinite mass, series convention)	$+0.0534\%$
R_H (reduced mass)	-0.0011%

The simple reduced-mass correction shrinks the error by ~**50**×. The remaining $+0.147$ meV residual (R_H is slightly under-bound relative to reality) is quantitatively explained below.

3.2 Lyman and Balmer spectral lines

Computed with R_H (reduced mass), $hc = 1239.84$ eV·nm; compared to reference values (Lyman in vacuum; Balmer historically quoted in air, converted to vacuum with $n_{\text{air}} = 1.000277$):

Series	Transition	λ_{model} (vacuum, nm)	$\lambda_{\text{reference}}$ (vacuum, nm)	Relative error
Lyman α	$2 \to 1$	121.568	121.567	0.0012%
Lyman β	$3 \to 1$	102.573	102.572	0.0013%
Lyman γ	$4 \to 1$	97.255	97.254	0.0008%
Balmer α (Hα)	$3 \to 2$	656.470	656.461	0.0014%
Balmer β (Hβ)	$4 \to 2$	486.274	486.270	0.0009%
Balmer γ (Hγ)	$5 \to 2$	434.173	434.167	0.0013%
Balmer δ (Hδ)	$6 \to 2$	410.293	410.288	0.0013%

All seven lines agree with experiment to within ~ 0.001% — remarkably uniform precision across both series, once the air-to-vacuum conversion is applied to the Balmer lines.

3.3 Quantifying the residual: the relativistic kinetic correction $m_{\text{eff}} = m/\sqrt{1 - V^2/c^2}$

The standard relativistic mass, $m_{\text{eff}}(V) = m/\sqrt{1 - V^2/c^2} = \gamma m$, inserted into the kinetic term of the energy functional $\langle H\rangle_{1s}(\beta) = \hbar^2/2m\beta^2 - Z_{eff}ke^2/\beta$, gives (full derivation in Corectie_Relativista_TauTimp_v2_RO.md):

$$\boxed{\Delta E_{\text{rel}} \approx -|E_0| \cdot \frac{1}{2}(Z_{eff}\alpha/n)^2} \qquad \text{(heuristic mass substitution)}$$

Compared with the **rigorous** relativistic term, already present in the extended TauTimp ondulatory equation (source document, §6.3, $\widehat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$, derived from the exact energy-momentum relation):

$$\boxed{\Delta E_{\text{rigorous}} = -\frac{T_0^2}{2mc^2} = -|E_0| \cdot \frac{1}{4}(Z_{eff}\alpha/n)^2} \qquad \Rightarrow \qquad \frac{\Delta E_{\text{rel}}}{\Delta E_{\text{rigorous}}} = 2.000 \text{ (exact)}$$

Applying both to hydrogen, starting from R_H (already reduced-mass-corrected):

Correction applied on top of R_H	ΔE (meV)	Resulting prediction (eV)	Gap vs. NIST (meV)
(none)	—	-13.598287	$+0.147$
ΔE_{rel} (heuristic $m/\sqrt{1 - V^2/c^2}$	-0.362	-13.598649	-0.215

Correction applied on top of R_H	ΔE (meV)	Resulting prediction (eV)	Gap vs. NIST (meV)
substitution)			
$\Delta E_{\text{rigorous}}$ $(-\hat{p}^4/8m^3c^2)$	-0.181	-13.598468	$\boxed{-0.034}$

The rigorous formula shrinks the gap by ~ 4.3 × (from 0.147 to 0.034 meV); the heuristic mass substitution, while having the correct physical sign (more binding, matching the Dirac formula), *overestimates* the correction by exactly a factor of 2 and actually **worsens** agreement with NIST for hydrogen. The final 0.034 meV residual is at the scale of the Lamb shift (4.375 μeV) and hyperfine structure, already formalized (fully or phenomenologically) in the source document (§8–9).

4 Helium — the contrast between the ion (one electron) and the neutral atom (two electrons)

4.1 He^+ — exact hydrogenic ion, no screening

$$E(\text{He}^+, 1s, \text{reduced mass}) = -54.415354 \text{ eV} \qquad I_2^{\text{NIST}}(\text{He}) = 54.41778 \text{ eV} \qquad \text{gap} = +2.427 \text{ meV}$$

Applying the same comparison as for hydrogen:

Correction applied	ΔE (meV)	Resulting gap (meV)
(none, reduced mass only)	—	$+2.427$
ΔE_{rel} (heuristic)	-5.795	-3.369 (worse)
$\Delta E_{\text{rigorous}}$	-2.897	$\boxed{-0.471}$ (~ 5 × better)

The same pattern as hydrogen: the rigorous formula significantly improves agreement; the heuristic mass substitution overestimates and worsens the prediction.

4.2 Neutral He — the screened Slater model

$$E_{\text{total}}(\text{He}, 1s^2, Z_{eff} = 1.70) = -78.6409 \text{ eV} \qquad -(I_1 + I_2)^{\text{NIST}} = -79.0052 \text{ eV}$$

$$\boxed{\text{Error} = -0.461\% \quad (3026 \text{ meV, absolute})}$$

The rigorous relativistic correction for this case ($\Delta E_{\text{rigorous}} = -1.513$ meV, next section) explains under 0.05% of this 3026 meV gap — going from one electron to two increases the error by **~2000**× relative to the available relativistic correction, the first direct manifestation of this document's central limitation: **the independent-electron approximation with a fixed effective charge cannot represent real electron–electron repulsion, and single-electron relativity cannot compensate for it.**

5 Lithium, Beryllium, Boron, Carbon — the systematic degradation of precision

5.1 Summary table, with the rigorous relativistic correction

Atom	Valence state	Z_{eff}	E_{model} (eV)	E^{NIST} (eV)	Relative error	$\Delta E_{\text{rigorous}}$ (meV)	Fraction of gap
H	$1s$	1	-13.5983	-13.5984	0.001%	-0.181	2.5%
He^+	$1s$	2	-54.4154	-54.4178	0.004%	-2.897	119.6%
He	$1s^2$	1.70	-78.6409	-79.0052	0.461%	-3.026	0.10%
Li	$2s^1$	1.30	-5.7484	-5.3917	6.62%	-0.032	0.009%
Be	$2s^2$	1.95	-12.9339	-9.3227	38.74%	-0.164	0.005%
B	$2p^1$	2.42	-19.9201	-8.2980	140.06%	-0.388	0.003%
C	$2p^2$	3.14	-33.5367	-11.2603	197.83%	-1.101	0.005%

(For H and He^+, "Fraction of gap" is computed against the residual remaining after the reduced-mass correction, not the raw Z_{eff}/R_∞ error; for the others, against the total $E_{\text{model}} - E^{\text{NIST}}$ gap.)

5.2 Physical explanation of the trend

The error grows **monotonically and nearly exponentially** with the number of electrons in the valence/near-valence ($n = 2$) shell. The cause: **the electron–electron repulsion energy (the Coulomb and exchange integrals of a full Hartree–Fock calculation) is not included** in the simple hydrogenic formula $E = -13.6057\, Z_{eff}^2/n^2$ — neither the reduced-mass nor the relativistic kinetic correction touches this cause, both being single-electron effects.

6 TauTimp vs. Schrödinger — the same error, by construction

For every case above, the cosmological tail $(r/R)^4$ of the modified TauTimp ondulatory equation remains, at all relevant equilibrium radii, between $\sim 10^{-132}$ and $\sim 10^{-134}$ — completely negligible, over 130 orders of magnitude below any error reported above, including the relativistic kinetic correction (μeV–meV). **None of the errors in the preceding sections can be attributed to a difference between TauTimp and Schrödinger** — both theories, at atomic scale, produce identical predictions, whether relativistically corrected or not.

7 Ways to improve the atomic model

7.1 For single-electron systems (H, He^+, hydrogenic ions)

1. **Reduced mass** — shrinks hydrogen's error by $\sim 50\times$ (Hydrogen section).
2. **The rigorous relativistic kinetic correction**, $\widehat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$ — shrinks the remaining residual by a further $\sim$ 4–5 $\times$ (Hydrogen, Helium sections). **Do not** use the heuristic mass substitution $m/\sqrt{1 - V^2/c^2}$ as a quantitative replacement — it has the correct sign but overestimates by exactly 2 $\times$ and worsens the prediction if applied alone.
3. **Spin-orbit coupling and the Darwin term** — already formalized in the extended TauTimp ondulatory equation (source, §6–7); complete the kinetic correction into a full Dirac formula.
4. **Lamb shift and hyperfine structure** — at the μeV scale, already treated (phenomenologically and fully, respectively) in the source (§8–9).

Fully applied, these corrections bring the prediction for single-electron systems into agreement with experiment limited only by the precision of the fundamental physical constants.

7.2 For multi-electron atoms (neutral He, Li, Be, B, C)

The corrections above **do not solve** the 0.46%–198% errors — as the Lithium–Carbon table clearly shows, the rigorous relativistic correction explains under 0.1% of any of these gaps. Effective directions:

5. **Self-consistent field (Hartree–Fock)** — replaces the fixed Z_{eff} with a self-consistently computed mean potential.
6. **Explicit electron correlation** (CI, Møller–Plesset, coupled-cluster CCSD(T), or Hylleraas for helium) — closes essentially all the gap remaining after Hartree–Fock.
7. **Antisymmetrized multi-electron wavefunction** (Slater determinant).
8. **Systematic, self-consistent relativistic corrections** for heavier atoms (Dirac–Fock) — useful only after correlation is solved, not before.

7.3 A specific extension of the TauTimp ondulatory equation

The TauTimp ondulatory equation, with or without the relativistic kinetic correction tested here, remains a **single-electron** equation. To surpass (not merely match) Schrödinger's precision for multi-electron atoms, TauTimp would need an explicit extension, currently absent from the source: coupling multiple ondulatory fields with electron–electron interaction terms, or a self-consistent Hartree–Fock-like loop built on the TauTimp formalism — remaining, like the first-principles derivation of the Lamb shift, an **open research program**.

8 Conclusions

9. **Schrödinger and TauTimp are experimentally indistinguishable at atomic scale** — even after adding the relativistic kinetic correction, the cosmological tail remains $\lesssim 10^{-130}$ relative.

10. **For single-electron systems** (H, He^+), the rigorous relativistic kinetic correction ($-\hat{p}^4/8m^3c^2$, already in the source document) shrinks the residual remaining after reduced mass by a factor of 4–5; the heuristic mass substitution $m/\sqrt{1 - V^2/c^2}$ has the correct physical sign but overestimates by exactly 2 $\times$ and should not be used quantitatively.

11. **For multi-electron atoms**, the relativistic correction — rigorous or heuristic — explains under 0.1% of the 0.46%–198% gaps against NIST, quantitatively confirming that single-electron relativity is not the missing mechanism.

12. **The most effective improvement path** for atoms B–C remains a self-consistent multi-electron treatment with explicit electron correlation — a methodological extension, not a change to the underlying wave theory.

9 Appendix: Experimental data (NIST) and constants used

Quantity	Value	Source
I_1(H)	13.598434 eV	NIST
I_1(He)	24.58741 eV	NIST
I_2(He)	54.41778 eV	NIST
I_1(Li)	5.3917 eV	NIST
I_1(Be)	9.3227 eV	NIST
I_1(B)	8.29803 eV	NIST
I_1(C)	11.26030 eV	NIST
Lyman α, β, γ (vacuum)	121.567; 102.572; 97.254 nm	standard spectroscopic data
Balmer $\alpha, \beta, \gamma, \delta$ (air)	656.279; 486.135; 434.047; 410.174 nm	standard spectroscopic data
n_{air} (visible)	1.000277	standard refractive index
R_∞	13.605693 eV	CODATA
R_H (reduced mass)	13.598287 eV	computed, $\mu = m_e m_p/(m_e + m_p)$
α (fine structure)	$7.2973525693 \times 10^{-3}$	CODATA
ΔE_{rel} (heuristic)	$-\lvert E_0\rvert (Z_{eff}\alpha/n)^2/2$	$m_{\text{eff}} = m/\sqrt{1 - V^2/c^2}$
$\Delta E_{\text{rigorous}}$	$-\lvert E_0\rvert (Z_{eff}\alpha/n)^2/4 = -T_0^2/2mc^2$	$\widehat{H}_{\text{rel}} = -\hat{p}^4/8m^3c^2$
R (TauTimp cosmological radius)	4.2355×10^{22} m	electron_TauTimp_final_RO.md

This document evaluates the shared precision of the Schrödinger equation and the TauTimp ondulatory equation against real experimental (NIST) data, for H, He, Li, Be, B, and C, now including an explicit quantification of the relativistic kinetic correction with effective mass $m/\sqrt{1 - V^2/c^2}$, compared against the rigorous relativistic term already present in the source document. The central conclusion stands: the two theories are indistinguishable at atomic scale; the real error for multi-electron atoms comes from the level of approximation of the model (fixed effective charge, no electron correlation), not from the choice of wave equation nor from single-electron relativistic corrections.This document is a synthesis, not a new derivation: every numerical value above is taken directly from, or algebraically derived from, electron_TauTimp_final_RO.md, Corectie_Relativista_TauTimp_v2_RO.md, Precizie_Schrodinger_TauTimp_EN.md, and the six per-element Heisenberg-equilibrium documents already present in this directory. It consolidates, in English, the percentage-difference conclusions scattered across that series into a single comparative reference for H, He, Li, Be, B, and C.

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